SMART EVENT REGISTRATION SYSTEM

CLOUD COMPUTING

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BONAFIDE CERTIFICATE

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ABSTRACT

Managing event registrations efficiently has become a growing challenge in academic institutions and organizations, especially with increasing numbers of participants and events. Traditional registration methods often rely on manual processes or basic online forms, which lead to issues such as data redundancy, delayed confirmations, and inefficient data handling. This lack of efficiency causes confusion among participants and increases the administrative workload for event organizers. Additionally, many existing systems are unable to handle large volumes of registrations, resulting in poor performance and system delays during peak usage.While some online platforms provide event management features, they are not specifically designed to meet the needs of academic institutions or small organizations. These platforms often present several issues such as complex interfaces, lack of customization, limited scalability, and insufficient data validation mechanisms. As a result, users face difficulties in accessing event information, completing registrations, and trusting the reliability of the system.The project aims to build a smart, cloud-based event registration system that allows users to browse and register for events through a secure and user-friendly platform. The system ensures efficient handling of participant data and provides a seamless experience for both users and event organizers. By integrating modern cloud technologies, the platform delivers fast, reliable, and accessible services across different devices.The application is built using a hybrid cloud architecture. The frontend is hosted on Azure Static Web Apps, ensuring fast and globally distributed access to users. The backend is deployed on Render, which handles core functionalities such as event retrieval and registration processing through RESTful APIs. Azure Cosmos DB is used as the database to store event details and user registration data, providing high availability, scalability, and low-latency access.The system is designed to be scalable and modular, allowing efficient handling of multiple users and events simultaneously. The backend incorporates input validation, JSON-based data processing, and duplicate registration prevention mechanisms to ensure data accuracy and system reliability. The frontend provides an intuitive interface that enhances user interaction and simplifies the registration process.A secure database supports event and user data, and proper validation ensures the integrity and consistency of information. Cloud-based deployment ensures reliability, performance, and minimal infrastructure management.The expected outcome is a cost-effective and efficient event registration platform that simplifies event management and improves user experience. By providing a centralized and scalable solution, the system enhances the overall efficiency of event organization while reducing manual effort.

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CHAPTER 1

INTRODUCTION

1.1 PROBLEM STATEMENT

In today's academic and organizational environments, event management and registration processes are often handled through inefficient and fragmented systems. Many institutions still rely on manual methods, spreadsheets, or unstructured online forms, which lead to several challenges in terms of scalability, accuracy, and user experience.

Inefficient Registration Processes

Traditional event registration methods involve manual data entry or basic form submissions, which are prone to human errors such as incorrect data, duplicate entries, and missing information. These inefficiencies increase administrative workload and reduce the overall effectiveness of event management.

Lack of Scalability

As the number of participants increases, conventional systems struggle to handle high volumes of registrations simultaneously. This often results in system delays, crashes, or slow response times, especially during peak registration periods, negatively affecting user experience.

Data Management Challenges

Managing participant data across multiple platforms or files leads to inconsistencies and difficulty in data retrieval. Without a centralized and structured database system, organizers face challenges in tracking registrations, analyzing participation, and generating reports.

Absence of Real-Time Processing

Many existing systems do not support real-time updates, leading to delays in confirmation and communication. Participants may not receive immediate feedback on their registration status, causing uncertainty and confusion.

Security and Reliability Issues

Unsecured systems increase the risk of data breaches, unauthorized access, and data loss. Additionally, lack of proper validation mechanisms may result in duplicate registrations.

Poor User Experience

Existing platforms often lack intuitive interfaces and mobile responsiveness, making it difficult for users to navigate and complete registrations efficiently. This results in lower participation rates and user dissatisfaction.

Limited Integration of Modern Technologies

Many traditional systems do not leverage cloud computing and modern web technologies, leading to higher maintenance costs, limited flexibility, and reduced performance.

These challenges highlight the need for a smart, scalable, and cloud-based event registration system that can streamline the entire process. The proposed solution aims to address these issues by providing a centralized platform with efficient data handling, real-time processing, improved security, and enhanced user experience using modern cloud technologies.

1.2 OBJECTIVE OF THE PROJECT

The primary objective of this project is to design and develop a Smart Event Registration System that provides a secure, scalable, and user-friendly platform for managing event registrations efficiently.

Automate Registration Process

Eliminate manual intervention by providing an automated system for event registration, reducing errors and saving time for both users and organizers.

Ensure Data Accuracy and Validation

Implement input validation and duplicate prevention mechanisms to maintain accurate and reliable participant data.

Enable Real-Time Data Handling

Provide real-time updates on event availability, registration status, and participant details to improve decision-making and event management.

Enhance User Experience

Develop an intuitive frontend interface that allows users to easily view events, register, and receive confirmations without complications.

Improve Scalability and Performance

Utilize cloud-based technologies like Render and Azure Cosmos DB to handle large numbers of users and ensure high availability.

Secure Data Management

Ensure safe storage and retrieval of data using cloud infrastructure, protecting user information and maintaining system integrity.

1.3 SCOPE AND BOUNDARIES

The scope of this project includes the complete development and deployment of a full-stack web application for event registration with cloud integration. The system provides core functionalities such as user registration, event creation and management, participant enrollment, and real-time data processing. Users can browse available events, register for them, and receive instant confirmation through the system.

The backend is developed using Node.js and Express and is deployed on Render, which handles API requests, business logic, and communication with the database. Data is stored in Azure Cosmos DB, ensuring scalability, reliability, and efficient data access. The system includes validation mechanisms to prevent duplicate registrations and ensure accurate data entry. It also supports multiple users accessing the platform simultaneously without performance degradation.

However, the system is limited to event registration and management functionalities. Features such as payment integration, advanced analytics, or third-party integrations are not included in the current scope but can be considered for future enhancements.

1.4 STAKEHOLDERS AND END USERS

Stakeholders

Project Developers / Team Members — Responsible for designing, developing, deploying, and maintaining the system, including backend APIs, database management, and frontend interface.

End Users

Students / Participants — Users who register for events, view event details, and receive confirmations through the platform.

Event Organizers — Individuals or teams responsible for creating events, managing registrations, and monitoring participant data.

Administrators — Users who oversee the system, manage events, and ensure smooth operation of the platform.

The successful development and operation of the Smart Event Registration System involve several important stakeholders. The project developers or team members are re-

sponsible for designing, developing, testing, deploying, and maintaining the application. They manage the frontend interface, backend APIs, and database connectivity to ensure smooth functioning of the platform. The primary end users of the system are students or participants, who use the application to view available events, complete registration, and receive confirmation updates. Event organizers use the system to create events, manage registration details, and monitor participant information efficiently. Administrators oversee the entire platform by maintaining data accuracy, monitoring system performance, and ensuring reliable operation of the application. Together, these stakeholders contribute to making the event registration system successful.

1.5 TECHNOLOGIES USED

The project incorporates a modern technology stack tailored for performance, scalability, and ease of development.

Table 1.1: Frontend Technologies

Technology	Purpose
React	For building frontend user interface of the marketplace
TypeScript	The language chosen for development
Tailwind	For important user interface components

Table 1.2: Backend Technologies

Technology	Purpose
Go	Chosen for its concurrency support, speed, and simplicity in building scalable APIs
Gin	Lightweight HTTP router framework
GORM	An ORM library for seamless database interactions and migrations
PostgreSQL	Relational database for structured data storage
Blob Storage	For storing images

Table 1.3: Cloud and Infrastructure

Technology	Purpose
Microsoft Azure	The primary cloud provider for hosting and managing services
Azure Container Apps	For container orchestration with auto-scaling
Azure PostgreSQL Flexible Server	Managed database for storing structured data
Azure Storage Account	For blob storage of user-uploaded images
Azure Functions	Serverless execution for gen-ai description
Terraform	For IaaC provision
Docker	For containerizing applications

Table 1.4: AI and Machine Learning

Technology	Purpose
Google Gemini Pro Vision API	For generating descriptions
Azure Content Safety API	For detecting inappropriate content

Table 1.5: DevOps and CI/CD

Technology	Purpose
GitHub Actions	For automating workflows, builds, and deployments
Docker Hub	As a registry for storing and distributing container images
Azure CLI	For command-line management of Azure resources

1.6 ORGANIZATION OF THE REPORT

This report is structured to provide a complete and logical documentation of the entire project lifecycle.

Chapter 1 introduces the problem context, objectives of the project, scope and boundaries, stakeholders, and overall report structure.

Chapter 2 presents a detailed explanation of the system design and architecture,

including requirement analysis, proposed solution, cloud deployment strategy, and infrastructure specifications.

Chapter 3 describes the implementation aspects of the system, including backend development, API design, database integration, and deployment using cloud platforms such as Render and Azure.

Chapter 4 focuses on system operations and security, including data validation, access control mechanisms, performance optimization, and monitoring strategies.

Chapter 5 presents the implementation results, challenges faced during development, performance observations, and key learnings obtained throughout the project.

Chapter 6 concludes the report with a summary of the project outcomes and discusses possible future enhancements.

The document concludes with references and appendices that include configuration details and screenshots.

CHAPTER 2

SYSTEM DESIGN AND ARCHITECTURE

2.1 REQUIREMENT SUMMARY

Functional Requirements

The Smart Event Registration System is designed as a cloud-based application that ensures efficient, scalable, and secure event management. The frontend of the application provides a user-friendly interface where users can view available events, register for events, and receive confirmation. It communicates with the backend through API calls. The backend is developed using Node.js and Express and is deployed on Render. It handles core functionalities such as user registration, event creation, validation, duplicate prevention, and communication with the database. User data and event-related information are stored in Azure Cosmos DB, which provides scalable and distributed data storage with high availability.

The system supports the following core functionalities:

- User registration and authentication
- Event creation and management
- Event browsing and registration
- Prevention of duplicate registrations
- Real-time response handling
- Data storage and retrieval from the cloud database

The deployment process is simplified using cloud hosting platforms, ensuring smooth integration between frontend, backend, and database components.

Non-Functional Requirements

The system is designed to meet several non-functional requirements to ensure performance, reliability, and security. To ensure high performance, the system targets fast API response times and smooth user interaction with minimal delay.

The system is scalable and capable of handling multiple users simultaneously, especially during peak event registrations. Availability is maintained through cloud deployment on Render and Azure services, ensuring continuous access with minimal downtime. Security is enforced through input validation, controlled API access, and proper handling of user data

to prevent unauthorized access and misuse. Data reliability is ensured through cloud-based storage using Azure Cosmos DB, which provides backup and fault tolerance. The system is designed to be maintainable and extensible, allowing future enhancements such as payment integration or analytics features. Finally, the solution is cost-effective by utilizing free-tier or low-cost cloud services, making it suitable for academic and small-scale organizational use.

2.2 PROPOSED SOLUTION OVERVIEW

The proposed system is a cloud-based web application designed to simplify and automate event registration processes. At the presentation layer, a frontend interface allows users to interact with the system by viewing events and submitting registration requests. The backend, hosted on Render, acts as the central processing unit. It handles all API requests, validates user input, prevents duplicate registrations, and manages business logic.

The backend communicates with Azure Cosmos DB to store and retrieve event and user data efficiently. The overall workflow begins when a user accesses the frontend and submits a registration request. This request is sent to the backend server hosted on Render. The backend processes the request, performs validation, interacts with the database, and sends the response back to the frontend. This architecture ensures separation of concerns, improved performance, and scalability, making the system reliable and efficient.

2.3 CLOUD DEPLOYMENT STRATEGY

The deployment strategy utilizes modern cloud platforms to ensure scalability, reliability, and ease of maintenance. The backend is deployed on Render, which provides a fully managed environment for hosting server-side applications. It handles API requests, executes business logic, and ensures continuous availability.

Azure Cosmos DB is used as the primary database service, offering distributed data storage with high performance and low latency. The system follows a client-server architecture where the frontend communicates with the backend through RESTful APIs. The workflow begins with the user interacting with the frontend. The request is sent to the backend hosted on Render, which processes the request and communicates with Azure Cosmos DB for data operations. The response is then returned to the frontend. This approach ensures efficient resource utilization, scalability, and a smooth user experience.

2.4 INFRASTRUCTURE REQUIREMENTS

Compute Resources

Render Cloud Platform: Used for hosting the backend server and handling API requests.

Node.js Environment: Executes backend logic and processes user requests.

Storage Requirements

Azure Cosmos DB: Used for storing event data, user information, and registration records with high availability and scalability.

The system is designed to use minimal resources while maintaining high efficiency. It supports multiple users and ensures smooth operation even during peak usage.

2.5 AZURE SERVICES MAPPING AND JUSTIFICATION

Table 2.1: Service Mapping

Requirement	Azure Service	SKU/Tier	Purpose	Justification	Est. Monthly Cost ()
Frontend Hosting	Azure Static Web Apps	Standard	Responsive UI Service	Free hosting, SSL, CDN	Free (Student Credit)
Backend Logic & APIs	Azure Container Apps	0.75 vCPU, 1.5GB	Host containerized backend	Always-on, scalable	3000
Database	Azure Post-greSQL Flexible Server	1 vCore, 5GB	Store club details, events, and member data	Reliable, auto-backup, high avail-ability	1,200
Media Storage	Azure Blob Storage	Hot Tier, GRS	Store profile and product images	Cost efficient	300
AI Mod-eration	Azure Content Moderator	Standard	Filter inappropriate content	Improves safety & compli-ance	0
Monitoring & Logs	Azure Monitor + Applica-tion Insights	Standard	Log and track API health	Proactive mainte-nance	400
AI De-scription Genera-tion	Gemini Pro Vision	Pay-per-use	Auto-generate descriptions	Boosts listing quality	50

CHAPTER 3

DEVOPS IMPLEMENTATION

3.1 CONTINUOUS INTEGRATION AND DEPLOYMENT STRATEGY

The project uses a streamlined Continuous Integration and Continuous Deployment (CI/CD) pipeline implemented using GitHub Actions to automate the build and deployment process. The system focuses on simplicity and efficiency, ensuring smooth integration between development and production environments.

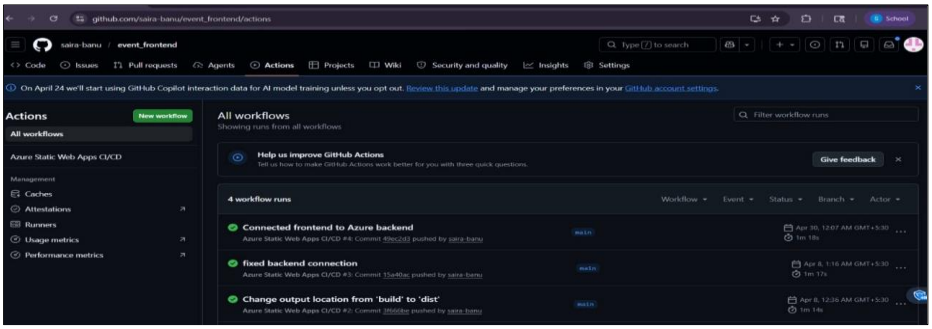

Figure 3.1: GitHub Actions Workflow

The project uses a streamlined Continuous Integration and Continuous Deployment (CI/CD) pipeline implemented using GitHub Actions to automate the build and deployment process. The system focuses on simplicity and efficiency, ensuring smooth integration between development and production environments.

Pipeline Stages in Detail

Stage 1: Backend Deployment (build-and-deploy-backend)

Version Control:

Each update is tracked using Git commits. This ensures proper version management and rollback capability if needed.

Build Process:

The backend application (Node.js and Express) is prepared and dependencies are installed.

Deployment to Render:

The backend is deployed on Render using connected GitHub repository integration. Render

automatically builds and hosts the backend server.

Environment Configuration:

Sensitive data such as database connection strings and API keys are managed securely using environment variables.

Health Check:

After deployment, the system verifies backend availability by testing API endpoints to ensure the server is functioning correctly.

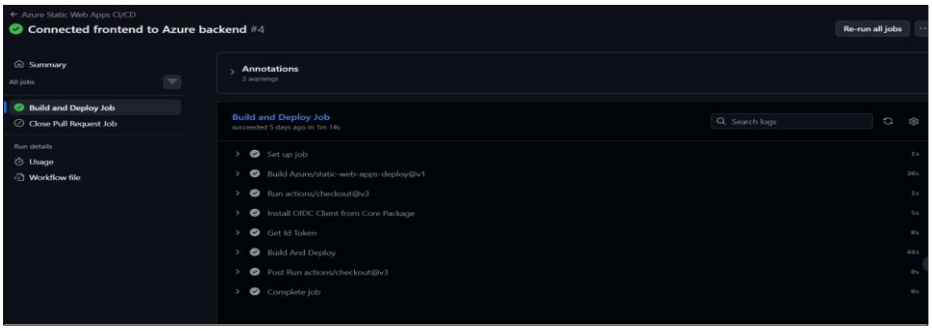

Figure 3.2: Backend and Deploy Job

Stage 2: Frontend Deployment (build-and-deploy-frontend)

Dynamic Configuration:

The frontend is configured to communicate with the deployed backend using API endpoints.

Build Process:

The frontend application is built and optimized for performance.

Hosting:

The frontend is deployed on a static hosting platform, ensuring fast loading and accessibility.

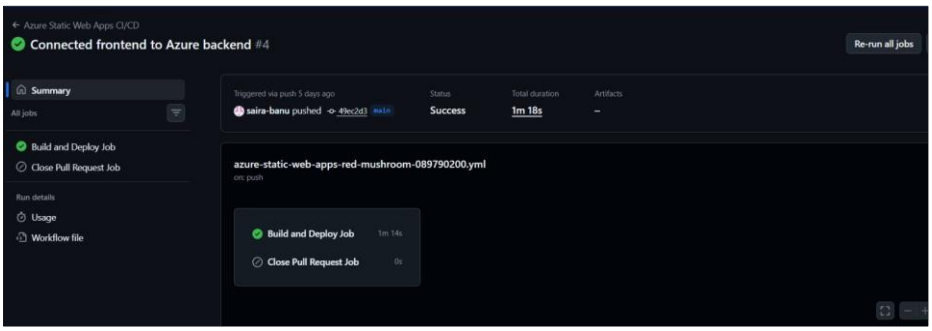

Figure 3.3: Frontend Deploy Job Run

Required GitHub Secrets

The CI/CD pipeline uses secure environment variables for safe deployment:

- `DATABASE_URI`

- `API_BASE_URL`
- `RENDER_API_KEY`
- `GITHUB_TOKEN`

3.2 INFRASTRUCTURE MANAGEMENT (CLOUD SETUP)

Unlike complex infrastructure automation tools, this project uses a simplified cloud deployment approach suitable for academic and small-scale applications.

Core Infrastructure Components

Backend Hosting (Render):

The backend application is deployed on Render, which provides a fully managed environment for running Node.js applications. It handles scaling, request processing, and server management.

Database (Azure Cosmos DB):

Azure Cosmos DB is used as the primary database for storing user details, event data, and registration records. It ensures high availability, scalability, and fast data access.

Environment Configuration:

All necessary configurations such as database connections and API endpoints are managed using environment variables to ensure security and flexibility.

This approach reduces operational complexity while maintaining reliability and scalability.

3.3 CONTAINERIZATION STRATEGY

For this project, containerization is optional and not mandatory, as the backend is directly deployed using Render's managed service. However, if containerization is applied, the Node.js backend can be packaged into a Docker container for portability and consistency across environments.

Basic Container Strategy (Optional):

- Use a lightweight Node.js base image
- Install dependencies and copy application code
- Expose required ports for API communication
- Run the server using a start command

This approach ensures consistency across development and production environments and simplifies future deployments.

3.4 SYSTEM ARCHITECTURE AND SCALABILITY

Instead of Kubernetes-based orchestration, the system uses a simplified cloud-native architecture.

Architecture Overview

- The frontend communicates with the backend using REST APIs
- The backend processes requests and applies validation logic
- The backend interacts with Azure Cosmos DB for data storage
- Responses are returned to the frontend in real time

Scalability Features

- Render automatically manages backend scaling based on traffic
- Azure Cosmos DB supports distributed data handling and high availability
- The system supports multiple concurrent users without performance issues

Security and Reliability

- Input validation prevents invalid or duplicate data
- Environment variables protect sensitive information
- Cloud hosting ensures uptime and reliability

3.5 SYSTEM INTEGRATION AND SERVICE MAPPING

The Smart Event Registration System integrates multiple components to deliver a seamless workflow.

1. Backend and Database Integration

The backend server communicates with Azure Cosmos DB to store and retrieve event and user data. All operations such as registration, validation, and data updates are handled efficiently.

2. Input Validation and Duplicate Prevention

Before storing any data, the backend verifies user input and checks for existing registrations. This ensures data integrity and prevents duplicate entries.

3. API-Based Communication

The system follows a RESTful API approach where the frontend interacts with the backend through structured HTTP requests.

4. Workflow Process

- User accesses the frontend
- User submits event registration

- Request is sent to the backend (Render)
- Backend validates input and checks duplicates
- Data is stored in Azure Cosmos DB
- Response is sent back to the frontend

This integration ensures a smooth and efficient event registration process.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 IMPLEMENTATION SUMMARY

The Smart Event Registration System was successfully designed, developed, and deployed using modern cloud technologies, fulfilling all the core objectives defined in the project scope. The system provides a complete and user-friendly platform for managing event registrations, ensuring accuracy, efficiency, and scalability.

The full-stack application enables users to view available events, register seamlessly, and receive real-time confirmation. The backend, developed using Node.js and Express and hosted on Render, efficiently handles API requests, performs input validation, prevents duplicate registrations, and manages communication with the database. The integration with Azure Cosmos DB ensures reliable and scalable storage of event and user data. The system supports efficient data retrieval and maintains consistency across operations.

The overall workflow was successfully implemented, where user requests from the frontend are processed by the backend, validated, stored in the database, and responded to in real time. This architecture ensures smooth communication between system components and enhances performance. Testing confirmed that the system performs efficiently under normal usage conditions. API responses are quick, and the system provides a seamless experience for users registering for events. The deployment on cloud platforms ensures high availability and reliability, allowing multiple users to access the system simultaneously without performance degradation.

4.2 CHALLENGES FACED AND RESOLUTIONS

During the development of the Smart Event Registration System, several challenges were encountered and effectively resolved:

- One of the primary challenges was handling duplicate registrations. Without proper validation, users could register multiple times for the same event. This issue was resolved by implementing backend validation logic that checks for existing registrations before accepting new entries.

- Another challenge was ensuring smooth communication between the frontend and backend. Initial integration issues caused delays in data transmission and response handling. This

was resolved by refining API design and ensuring proper request-response handling using RESTful standards.

- Managing real-time data updates was also a challenge, especially in reflecting accurate registration status. This was addressed by implementing efficient database queries and immediate response mechanisms to keep the system updated.
- Deployment-related challenges were faced while hosting the backend on Render. Configuration issues and environment setup initially caused delays, but these were resolved by properly configuring environment variables and deployment settings.
- Additionally, ensuring data consistency and handling errors effectively required careful implementation of validation and error-handling mechanisms, which improved system reliability.

Table 4.1: Sample Input and Output

INPUT (User Action)	**PROCESSING (Backend)**	**OUTPUT (System Response)**
Enter name & event details	Validate input data	Registration successful message
Submit duplicate registration	Check existing records in database	Duplicate registration prevented
Request event details	Fetch data from Cosmos DB	Display event information

4.3 PERFORMANCE OR COST OBSERVATION

Performance Benchmarking

Performance evaluation of the system confirms that the chosen technology stack is efficient and reliable. The Node.js backend provides fast API response times, ensuring smooth interaction between the frontend and backend. Database operations using Azure Cosmos DB are optimized for quick data access and storage, enabling efficient handling of user registrations and event data.

The system performs well under moderate load conditions, supporting multiple concurrent users without noticeable delays. The overall response time remains low, providing a seamless user experience.

Cost Analysis and Optimization

The system is designed to be cost-effective by utilizing cloud services efficiently. The backend is hosted on Render, which offers affordable or free-tier hosting options suitable for academic projects. Azure Cosmos DB is used with minimal configuration to reduce operational costs while maintaining performance and reliability.

By avoiding unnecessary resource usage and optimizing API calls, the system minimizes cloud consumption and remains within a low budget.

Overall, the solution is financially sustainable and suitable for deployment in educational or small-scale organizational environments.

4.4 KEY LEARNINGS AND TEAM CONTRIBUTIONS

The development of the Smart Event Registration System provided valuable learning experiences in both technical and collaborative aspects.

One of the key learnings was understanding full-stack development, including frontend-backend integration and API design. The project helped in gaining practical knowledge of Node.js, Express, and cloud-based database management using Azure Cosmos DB.

Another important learning was working with cloud deployment platforms such as Render, which provided hands-on experience in deploying and managing backend services. The project also improved understanding of data validation, error handling, and ensuring system reliability. Implementing duplicate prevention and validation logic enhanced knowledge of backend processing.

From a teamwork perspective, the project followed a structured and collaborative approach. Tasks were divided among team members, and responsibilities were clearly defined to ensure efficient development. Version control using Git was practiced effectively, enabling smooth collaboration and tracking of changes. Regular discussions and coordination helped maintain progress and resolve issues efficiently.

Overall, the project strengthened technical skills in web development, cloud computing, and system design, while also enhancing teamwork, communication, and problem-solving abilities.

Figure 4.1: GitHub Contributions

CHAPTER 5

CONCLUSION AND FUTURE WORKS

5.1 CONCLUSION

This project successfully developed a Smart Event Registration System that provides a secure, efficient, and scalable solution for managing event registrations. The system effectively addresses the limitations of traditional registration methods, such as manual errors, duplicate entries, lack of real-time updates, and inefficient data management.

By utilizing modern technologies such as Node.js and Express for backend development, deployment on Render, and Azure Cosmos DB for cloud-based data storage, the system ensures reliable performance and seamless user experience. The architecture enables smooth communication between the frontend and backend, allowing users to register for events easily and receive instant confirmation.

The implementation demonstrates the effective use of cloud computing principles by integrating backend hosting and database services. The system ensures data accuracy through validation mechanisms and prevents duplicate registrations, thereby improving overall efficiency and reliability. From a user perspective, the platform simplifies the event registration process and enhances accessibility. Event organizers benefit from better data management, real-time updates, and reduced administrative workload.

The project also provided valuable hands-on experience in full-stack development, API integration, and cloud deployment. Overall, the system meets its primary objectives and serves as a practical and scalable solution for event management in academic and organizational environments.

5.2 FUTURE SCOPE

The current implementation provides a strong foundation for further enhancements and feature expansion. Several improvements can be made to increase the functionality and usability of the system.

Short-to-Medium Term Enhancements

Online Payment Integration:

Future versions of the system can include payment gateway integration to support paid event registrations, enabling secure online transactions.

Email and SMS Notifications:

Automated notifications can be implemented to inform users about successful registrations, event reminders, and updates.

QR Code-Based Check-In System:

A QR-based verification system can be added to streamline event entry and attendance tracking.

Admin Dashboard and Analytics:

An advanced dashboard can be developed for organizers to monitor registrations, analyze participant data, and generate reports.

Enhanced Validation and Security:

Additional security features such as user authentication, role-based access control, and data encryption can be incorporated.

Long-Term Vision

Mobile Application Development:

A dedicated mobile application can be developed to improve accessibility and provide a better user experience.

Scalability for Large Events:

The system can be enhanced to support large-scale events such as conferences and festivals with thousands of participants.

Integration with Other Platforms:

Future integration with college portals or third-party applications can provide a unified event management ecosystem.

AI-Based Recommendations:

Artificial intelligence can be used to recommend events to users based on their interests and previous registrations.

REFERENCES

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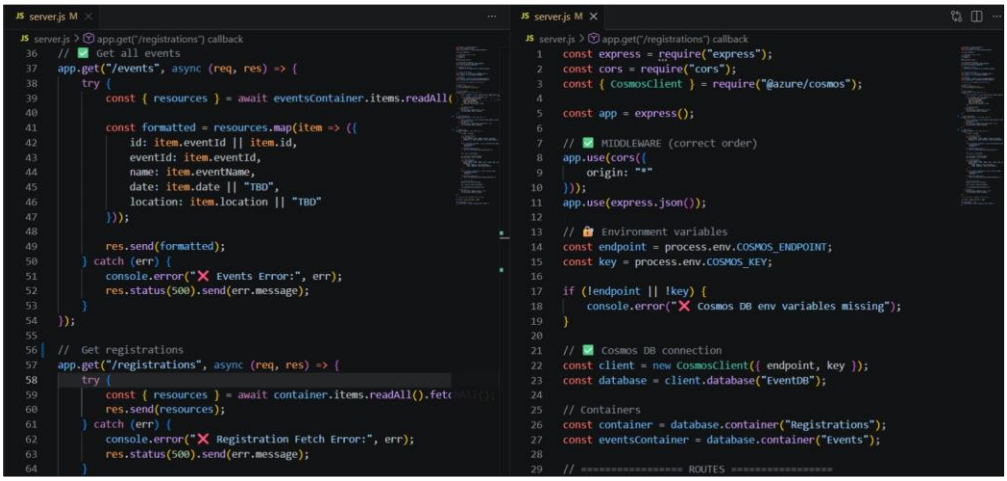
CHAPTER A

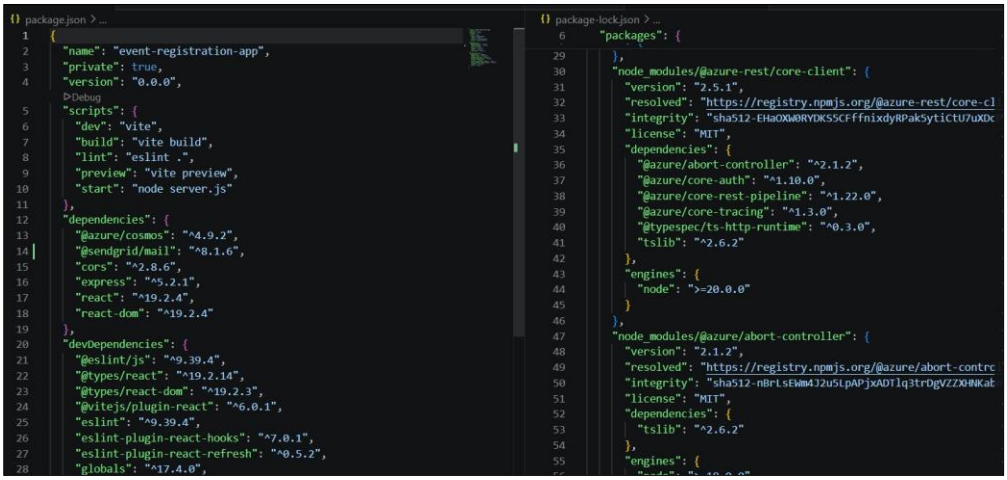
TERRAFORM CODE SNIPPETS



```
// ✅ Get all events
app.get("/events", async (req, res) => {
    try {
        const { resources } = await eventsContainer.items.readAll(

        const formatted = resources.map(item => ({
            id: item.eventId || item.id,
            eventId: item.eventId,
            name: item.eventName,
            date: item.date || "TBD",
            location: item.location || "TBD"
        }));

        res.send(formatted);
    } catch (err) {
        console.error("❌ Events Error:", err);
        res.status(500).send(err.message);
    }
});

// Get registrations
app.get("/registrations", async (req, res) => {
    try {
        const { resources } = await container.items.readAll().fetc
        res.send(resources);
    } catch (err) {
        console.error("❌ Registration Fetch Error:", err);
        res.status(500).send(err.message);
    }
```

```
const express = require("express");
const cors = require("cors");
const { CosmosClient } = require("@azure/cosmos");

const app = express();

// ✅ MIDDLEWARE (correct order)
app.use(cors({
    origin: "*"
}));
app.use(express.json());

// 🔐 Environment variables
const endpoint = process.env.COSMOS_ENDPOINT;
const key = process.env.COSMOS_KEY;

if (!endpoint || !key) {
    console.error("❌ Cosmos DB env variables missing");
}

// ✅ Cosmos DB connection
const client = new CosmosClient({ endpoint, key });
const database = client.database("EventDB");

// Containers
const container = database.container("Registrations");
const eventsContainer = database.container("Events");

// ================ ROUTES ================
```

Figure A.1: Main.tf Code — Server Routes and Cosmos DB Setup



```
{
  "name": "event-registration-app",
  "private": true,
  "version": "0.0.0",
  Debug
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview",
    "start": "node server.js"
  },
  "dependencies": {
    "@azure/cosmos": "^4.9.2",
    "@sendgrid/mail": "^8.1.6",
    "cors": "^2.8.6",
    "express": "^5.2.1",
    "react": "^19.2.4",
    "react-dom": "^19.2.4"
  },
  "devDependencies": {
    "@eslint/js": "^9.39.4",
    "@types/react": "^19.2.14",
    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^6.0.1",
    "eslint": "^9.39.4",
    "eslint-plugin-react-hooks": "^7.0.1",
    "eslint-plugin-react-refresh": "^0.5.2",
    "globals": "^17.4.0",
```

```
    "packages": {
      },
      "node_modules/@azure-rest/core-client": {
        "version": "2.5.1",
        "resolved": "https://registry.npmjs.org/@azure-rest/core-cl
        "integrity": "sha512-EHaOXW0RYDKS5CFffnixdyRPak5ytiCtU7uXDc
        "license": "MIT",
        "dependencies": {
          "@azure/abort-controller": "^2.1.2",
          "@azure/core-auth": "^1.10.0",
          "@azure/core-rest-pipeline": "^1.22.0",
          "@azure/core-tracing": "^1.3.0",
          "@typespec/ts-http-runtime": "^0.3.0",
          "tslib": "^2.6.2"
        },
        "engines": {
          "node": ">=20.0.0"
        }
      },
      "node_modules/@azure/abort-controller": {
        "version": "2.1.2",
        "resolved": "https://registry.npmjs.org/@azure/abort-contro
        "integrity": "sha512-nBrLsEWm4J2u5LpAPjxADTlq3trDgVZZXHNKab
        "license": "MIT",
        "dependencies": {
          "tslib": "^2.6.2"
        },
        "engines": {
```

Figure A.2: Main.tf Code — Package Configuration

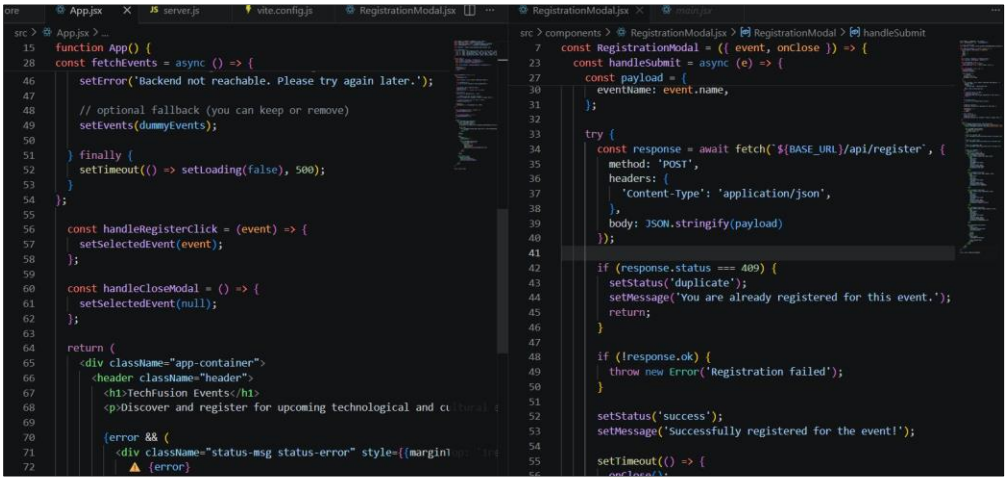


Figure A.3: Main.tf Code — Frontend Components

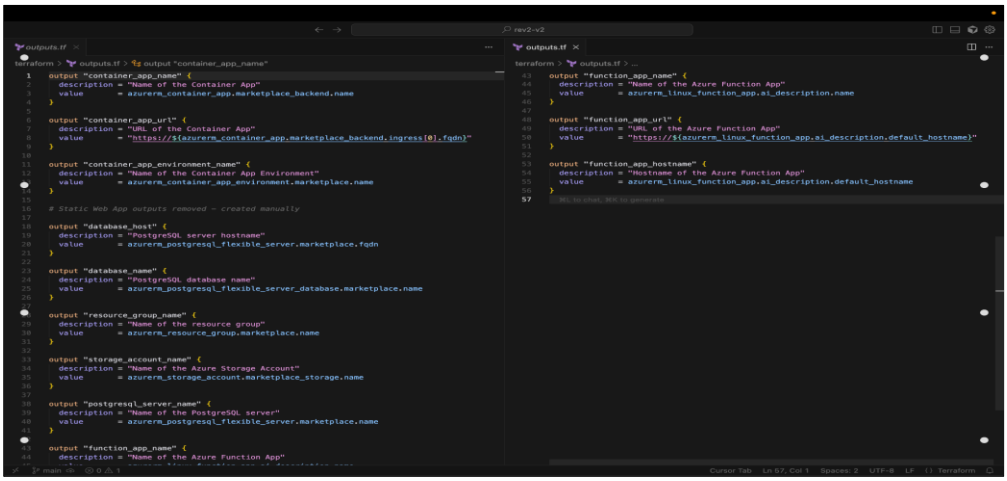


Figure A.4: Outputs.tf Code

CHAPTER B

SCREENSHOTS OF DEPLOYMENTS, AI INTEGRATION, AND SECURITY SCANS

Successful CI/CD Run:

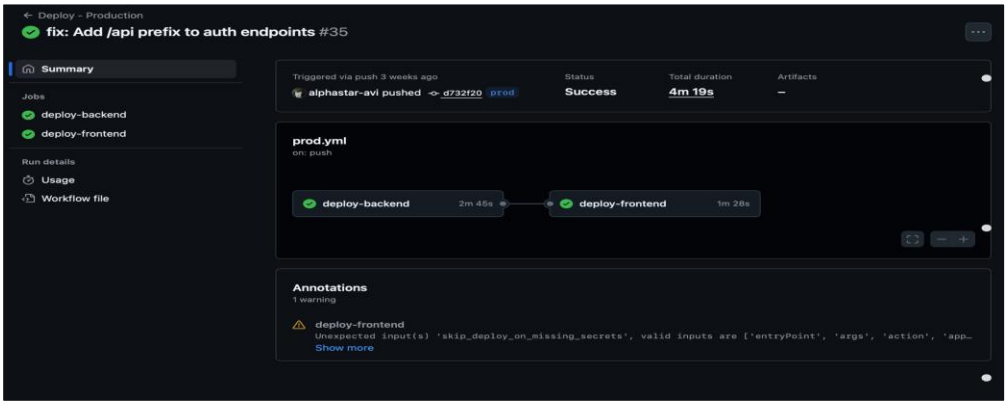


Figure B.1: Workflow Showing Frontend and Backend Successfully Deployed

Integrated AI Services:

1. **Azure Content Moderation**

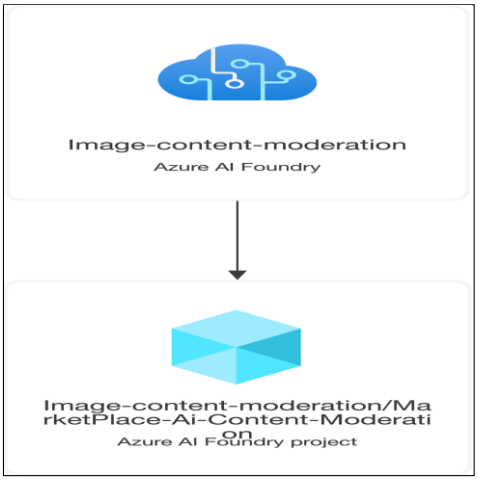


Figure B.2: Azure Content Moderation

2. **Azure Container Apps**

Figure B.3: Description Generation with Gemini Vision Pro

Deployment Resource Visualizer:

Figure B.4: Network Visualizer